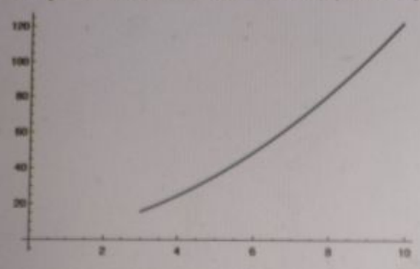
File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

+ Insert Cell...

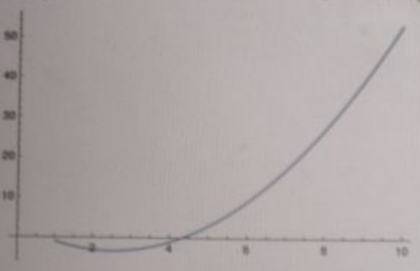
In[•]:=
```
Plot[x^2+2x+1, {x, 3, 10}, AxesOrigin → {0, 0}]
```

Out[•]=



In[•]:=
```
Plot[x^2-5x+3, {x, 1, 10}, AxesOrigin → {0, 0}]
```

Out[•]=



In[•]:=
```
Mesh → All
```

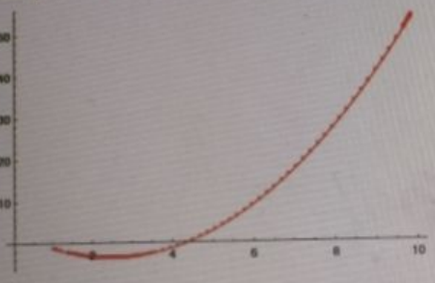
File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

+ Insert Cell...

Out[•]=
```
Mesh → 10.1235
```

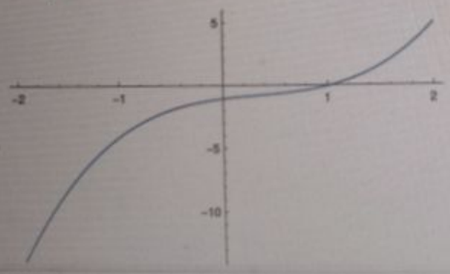
In[•]:=
```
Plot[x^2-5x+3, {x, 1, 10}, AxesOrigin → {0, 0}, Mesh → All, PlotStyle → Pink]
```

Out[•]=



In[•]:=
```
Plot[x^3-x^2+x-1, {x, -2, 2}]
```

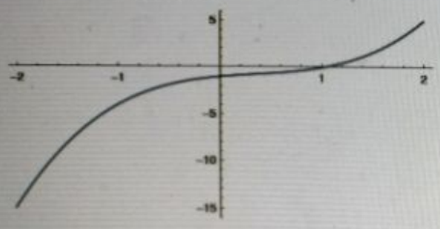
Out[•]=



+ Insert Cell...

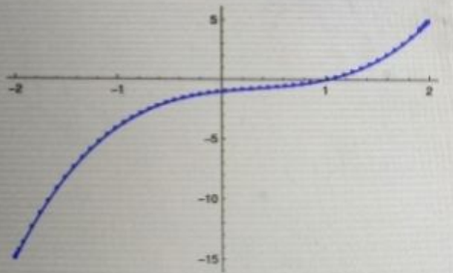
In[•]:=
```
Plot[x^3-x^2+x-1, {x, -2, 2}, AspectRatio → 1/2]
```

Out[•]=



In[•]:=
```
Plot[x^3-x^2+x-1, {x, -2, 2}, Mesh → All, PlotStyle → Blue]
```

Out[•]=



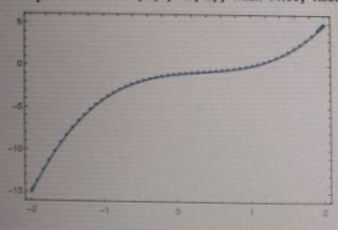
In[•]:=
```
Plot[x^3-x^2+x-1, {x, -2, 2}, Mesh → All, Axes → False]
```

```
In[•]:= Plot[x^3-x^2+x-1, {x, -2, 2}, Mesh → All, Axes → False]
```

Out[•]=

```
In[•]:= Plot[x^3-x^2+x-1, {x, -2, 2}, Mesh → All, Axes → False, Frame → True]
```

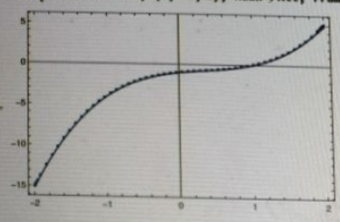
Out[•]=




```
In[•]:= Plot[x^3-x^2+x-1, {x, -2, 2}, Mesh → All, Frame → True]
```

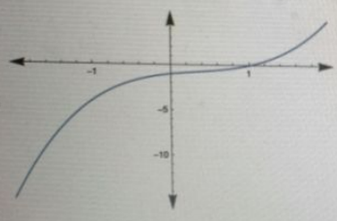
```
In[•]:= Plot[x^3-x^2+x-1, {x, -2, 2}, Mesh → All, Frame → True]
```

Out[•]=




```
In[•]:= Plot[x^3-x^2+x-1, {x, -2, 2}, AxesStyle → Arrowheads[{-0.05, 0.05}]]
```

Out[•]=

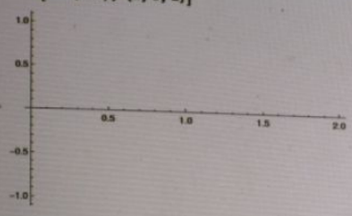



```
In[•]:= Plot[sin (x^2), {x, 0, 2}]
```

File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

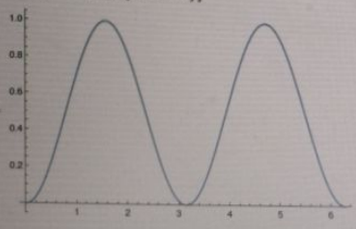
```
In[•]:= Plot[sin (x^2), {x, 0, 2}]
```

Out[•]=




```
In[•]:= Plot[Sin[x]^2, {x, 0, 2 π}]
```

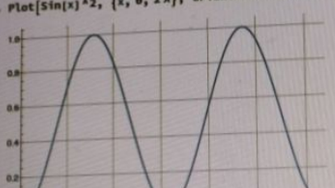
Out[•]=




```
In[•]:= Plot[Sin[x]^2, {x, 0, 2 π}, GridLines → Automatic]
```

File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

```
In[•]:= Plot[Sin[x]^2, {x, 0, 2 π}, GridLines → Automatic]
```




```
In[•]:= Manipulate[Plot[x + sin[1 / x], {x, 0, r}] (r^2, 0.1, 2)]
```

Out[•]=

$$\left\{r^2\,\text{Plot}\left[x+\sin\left[\frac{1}{x}\right],\{x,0,r\}\right],\ 0.1\,\text{Plot}\left[x+\sin\left[\frac{1}{x}\right],\{x,0,r\}\right],\ 2\,\text{Plot}\left[x+\sin\left[\frac{1}{x}\right],\{x,0,r\}\right]\right\}$$

Plot: Limiting value r in {x, 0, r} is not a machine-sized real number.

Plot: Limiting value r in {x, 0, r} is not a machine-sized real number.

Plot: Limiting value r in {x, 0, r} is not a machine-sized real number.

Plot: Limiting value r in {x, 0, r} is not a machine-sized real number.

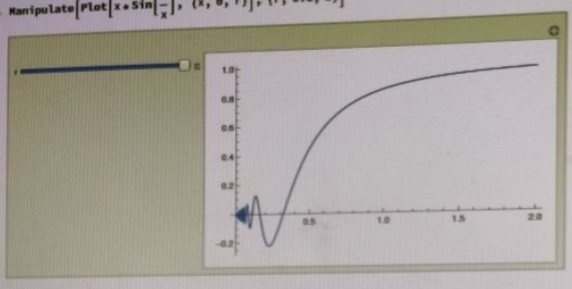
Plot: Limiting value r in {x, 0, r} is not a machine-sized real number.

File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

+ Insert Cell...

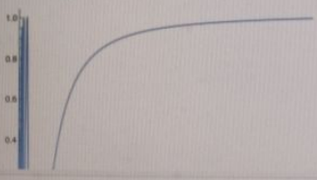
In[-]:= Manipulate[Plot[x * Sin[1/x], {x, 0, r}], {r, 0.1, 2}]

Out[-]=



In[-]:= Plot[Cos[1/x], {x, 0, 2 π}]

Out[-]=

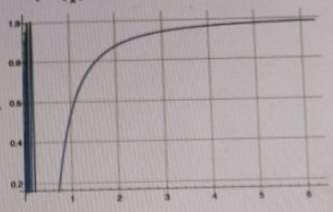


File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

+ Insert Cell...

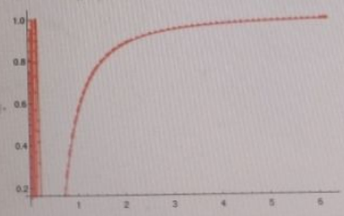
In[-]:= Plot[Cos[1/x], {x, 0, 2 π}, GridLines → Automatic]

Out[-]=



In[-]:= Plot[Cos[1/x], {x, 0, 2 π}, Mesh → All, PlotStyle → Pink]

Out[-]=

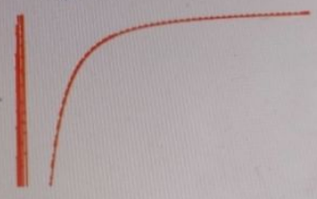


File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

+ Insert Cell...

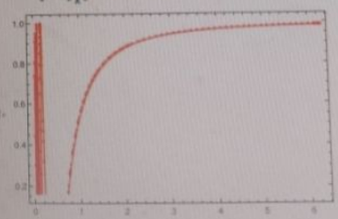
In[-]:= Plot[Cos[1/x], {x, 0, 2 π}, Mesh → All, PlotStyle → Pink, Axes → False]

Out[-]=



In[-]:= Plot[Cos[1/x], {x, 0, 2 π}, Mesh → All, PlotStyle → Pink, Frame → True]

Out[-]=

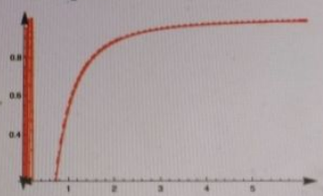


File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

+ Insert Cell...

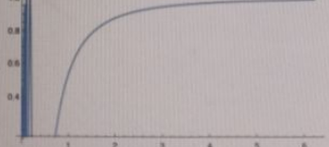
In[-]:= Plot[Cos[1/x], {x, 0, 2 π}, Mesh → All, PlotStyle → Pink, AxesStyle → Arrowheads[{-0.05, 0.05}]]

Out[-]=



In[-]:= Plot[Cos[1/x], {x, 0, 2 π}, AspectRatio → 1 / 2]

Out[-]=

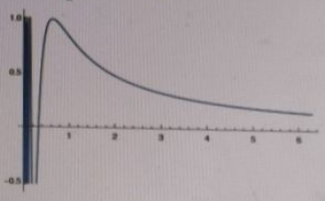


In[-]:= Plot[Sin[1/x], {x, 0, 2 π}]

File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

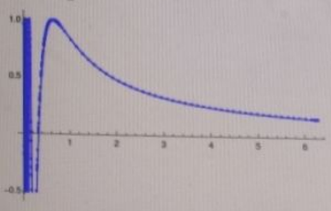
In[-]:=
```
Plot[Sin[1/x], {x, 0, 2 π}]
```

Out[-]=



In[-]:=
```
Plot[Sin[1/x], {x, 0, 2 π}, Mesh → All, PlotStyle → Blue]
```

Out[-]=

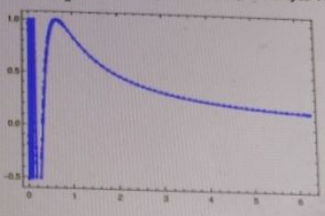


In[-]:=
```
Plot[Sin[1/x], {x, 0, 2 π}, Mesh → All, PlotStyle → Blue, Axes → False, Frame → True]
```

File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

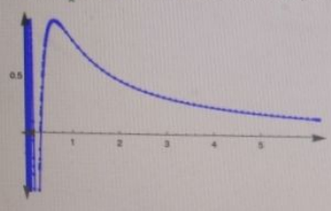
In[-]:=
```
Plot[Sin[1/x], {x, 0, 2 π}, Mesh → All, PlotStyle → Blue, Axes → False, Frame → True]
```

Out[-]=



In[-]:=
```
Plot[Sin[1/x], {x, 0, 2 π}, Mesh → All, PlotStyle → Blue, AxesStyle → Arrowheads[{-0.05, 0.05}]]
```

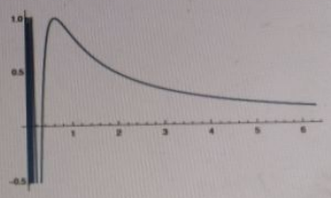
Out[-]=



File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

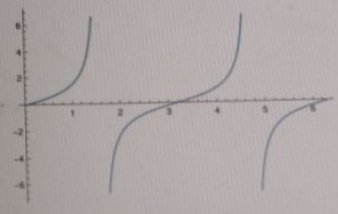
In[-]:=
```
Plot[Sin[1/x], {x, 0, 2 π}, AxesOrigin → {0, 0}]
```

Out[-]=



In[-]:=
```
Plot[Tan[x], {x, 0, 2 π}]
```

Out[-]=

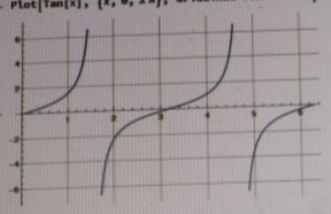


In[-]:=
```
Plot[Tan[x], {x, 0, 2 π}, GridLines → Automatic]
```

File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

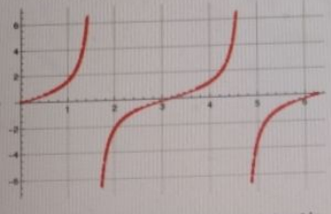
In[-]:=
```
Plot[Tan[x], {x, 0, 2 π}, GridLines → Automatic]
```

Out[-]=



In[-]:=
```
Plot[Tan[x], {x, 0, 2 π}, GridLines → Automatic, Mesh → All, PlotStyle → Red]
```

Out[-]=

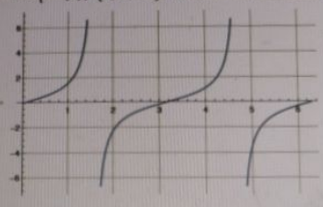


In[-]:=
```
Plot[Tan[x], {x, 0, 2 π}, GridLines → Automatic, AxesOrigin → {0, 0}]
```

File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

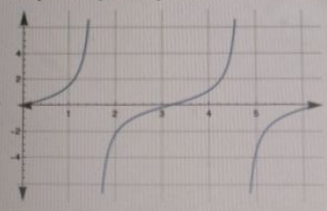
In[-]:=
```
Plot[Tan[x], {x, 0, 2 π}, GridLines → Automatic, AxesOrigin → {0, 0}]
```

Out[-]=



In[-]:=
```
Plot[Tan[x], {x, 0, 2 π}, GridLines → Automatic, AxesStyle → Arrowheads[{-0.05, 0.05}]]
```

Out[-]=



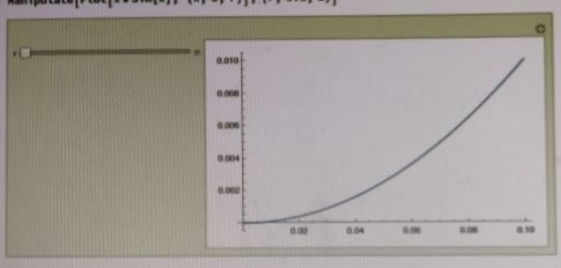
In[-]:=
```
Manipulate[Plot[x * Sin[x], {x, 0, r}], {r, 0.1, 2}]
```

MATHEMATICA.nb * - Wolfram Mathematica 12.2

File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

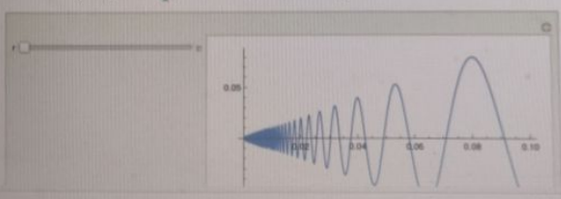
In[-]:=
```
Manipulate[Plot[x * Sin[x], {x, 0, r}], {r, 0.1, 2}]
```

Out[-]=



In[-]:=
```
Manipulate[Plot[x * Cos[1/x], {x, 0, r}], {r, 0.1, 2}]
```

Out[-]=



File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

In[-]:=
```
f[x_] = x^2 - 3 x + 1
f[x_] := x^2 - 3 x + 1
Table[f[x], {x, -10, 10}]
```

Set: Tag Times in f x_ is Protected.

Out[-]= $1 - 3x + x^2$

Out[-]= {131, 109, 89, 71, 55, 41, 29, 19, 11, 5, 1, -1, -1, 1, 5, 11, 19, 29, 41, 55, 71}

In[-]:=
```
f[x_] := x^2 - 3 x + 1
Table[f[x], {x, 1, 2, 5}]
```

Out[-]= {-1}

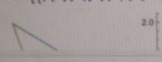
In[-]:=
```
f[x_] := x^2 - 3 x + 1
Griddata = {{x, f[x]}, {x, -1, 5}}
```

In[-]:=
```
f[x_] := x^2 - 3 x + 1
Grid[data] = {{x, f[x]}, {x, -5, 5}}
```

In[-]:=
```
TextGrid[Table Contents, Alignment → Right, Dividers → {Center, {False, True}}, Spacings → 2]
```

Out[-]= Grid[Contents Table, Alignment → Right, Dividers → {Center, {False, True}}, Spacings → 2]

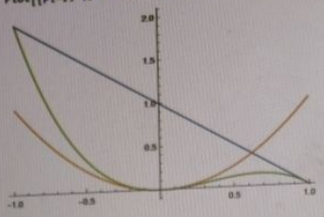
In[-]:=
```
p[x_] := 1 - x
q[x_] := x^2
Plot[{p[x], q[x], p[x] + q[x]}, {x, -1, 1}]
```



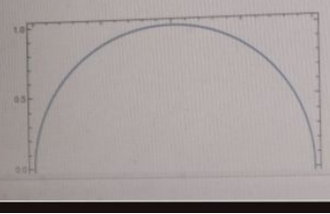
File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

In[-]:=
```
p[x_] := 1 - x
q[x_] := x^2
Plot[{p[x], q[x], p[x] + q[x]}, {x, -1, 1}]
```

Out[-]=

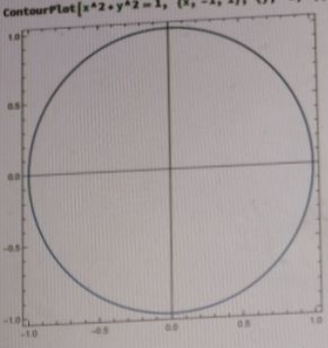


In[-]:=
```
ContourPlot[x^2 + y^2 == 1, {x, -1, 1}, {y, -1, 1}]
```

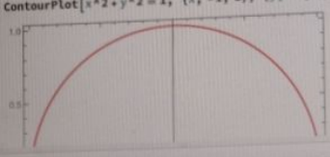


```
In[-]:= ContourPlot[x^2 + y^2 == 1, {x, -1, 1}, {y, -1, 1}, Axes → True]
```

Out[-]=

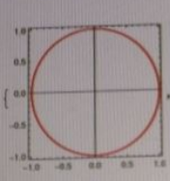


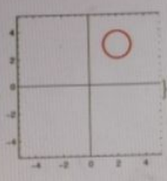
```
In[-]:= ContourPlot[x^2 + y^2 == 1, {x, -1, 1}, {y, -1, 1}, Axes → True, ContourStyle → Red]
```



```
In[-]:= {ContourPlot[x^2 + y^2 == 1, {x, -1, 1}, {y, -1, 1}, Axes → True, ContourStyle → Red],
         ContourPlot[(x - 2)^2 + (y - 3)^2 == 1, {x, -5, 5}, {y, -5, 5}, Axes → True, ContourStyle → Red]}
```

Out[-]= {



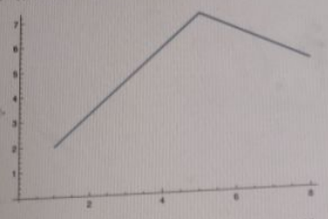


}

```
In[-]:= data {{1, 2}, {5, 7}, {8, 5}, {8, 4}, {5, 7}, {8, 5}, {8, 4}, {1, 2}, {5, 7}}
Out[-]= {{data, 2 data}, {5 data, 7 data}, {8 data, 5 data}, {8 data, 4 data}, {5 data, 7 data}, {8 data, 5 data}, {8 data, 4 data}, {data, 2 data}, {5 data, 7

In[-]:= Plotdata {{1, 2}, {5, 7}, {8, 5}, {8, 4}, {5, 7}, {8, 5}, {8, 4}, {1, 2}, {5, 7}}
Out[-]= {{Plotdata, 2 Plotdata}, {5 Plotdata, 7 Plotdata}, {8 Plotdata, 5 Plotdata}, {8 Plotdata, 4 Plotdata},
         {5 Plotdata, 7 Plotdata}, {8 Plotdata, 5 Plotdata}, {8 Plotdata, 4 Plotdata}, {Plotdata, 2 Plotdata}, {5 Plotdata, 7 Plotdata}}

In[-]:= data
        Plotdata {{1, 2}, {5, 7}, {8, 5}, {8, 4}}
Out[-]= data
Out[-]= {{Plotdata, 2 Plotdata}, {5 Plotdata, 7 Plotdata}, {8 Plotdata, 5 Plotdata}, {8 Plotdata, 4 Plotdata}}
```

```
In[-]:= data
        Plotdata {{1, 2}, {5, 7}, {8, 5}, {8, 4}}
Out[-]= data
Out[-]= {{Plotdata, 2 Plotdata}, {5 Plotdata, 7 Plotdata}, {8 Plotdata, 5 Plotdata}, {8 Plotdata, 4 Plotdata}}

In[-]:= data = {{1, 2}, {5, 7}, {8, 5}}
        ListPlot[data, Joined → True]
Out[-]= {{1, 2}, {5, 7}, {8, 5}}
```

Out[-]=



```
In[-]:= Show[scatterplot, Plot[x^2 - 2 x + 1, {x, -2, 2}]]
```

Show: Could not combine the graphics objects in Show[scatterplot,

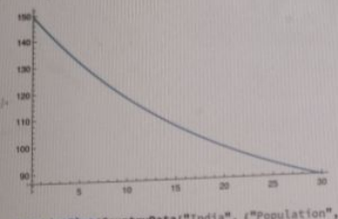
```
In[-]:= Show[scatterplot, Plot[x^2 - 2 x + 1, {x, -2, 2}]]
```

Show: Could not combine the graphics objects in Show[scatterplot,

Out[-]= Show[scatterplot,

```
In[-]:= Plot[69.348 + 80.148 * 0.95015^x, {x, 0, 30}]
```

Out[-]=



```
In[-]:= DataListPlot[CountryData["India", {"Population", (1925, 2025)}]]
```

```
In[•]:= DataListPlot[CountryData["India", ("Population", (1925, 2025))]]
```

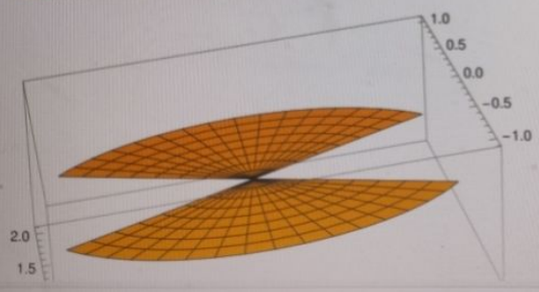
Out[•]= DataListPlot[TimeSeries[Time: 01 Jan 1925 GMT+5.5 to 01 Jan 2023 GMT+5.5 Data points: 99]]

Out[•]= DataListPlot[TimeSeries[Time: 01 Jan 1925 GMT+5.5 to 01 Jan 2023 GMT+5.5 Data points: 99]]

```
In[•]:= RevolutionPlot3D[x^2/3+y^2/3 = 1, {x, -1, 1}, {y, -1, 1}]
```

Set: Tag Plus in 0.333238 + 0.333238 is Protected.

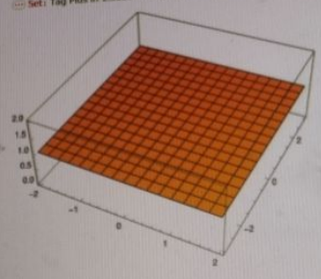
Out[•]=




```
In[•]:= Plot3D[x^2/3+y^2/3 = 1, {x, -2, 2}, {y, -3, 3}]
```

Set: Tag Plus in 1.33295 + 2.99914 is Protected.

Out[•]=




```
In[•]:= NSolve[x^2+2x+1 = 0, x, 40]
```

Set: Tag Plus in $1+2x+x^2$ is Protected.

```
In[•]:= Grid[Solve[x^2+2x+1 = 0, x] Alignment → Right]
```

Set: Tag Plus in $1+2x+x^2$ is Protected.

Solve: 0 is not a quantified system of equations and inequalities.

Solve: 0 is not a quantified system of equations and inequalities.

Out[•]= Grid[Alignment Solve[0, x] → Right]

```
In[•]:= Simplify[(x^3-1)/(x-1)]
```

Out[•]= $1+x+x^2$

```
In[•]:= Solve[{x+y-9 = 0, x-y = 0}, {x, y}]
```

Set: Tag Plus in $-9+x+y$ is Protected.

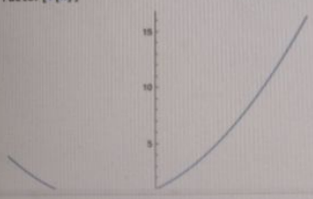
Set: Tag Plus in $x-y$ is Protected.

Solve: 0.646 0 is not a quantified system of equations and inequalities.

Out[•]= Solve[{0, 0}, {x, y}]

```
In[•]:= f[x_] := (x^2+2x+1)
        Plot[f[x], {x, -3, 3}]
        Factor[f[x]]
```

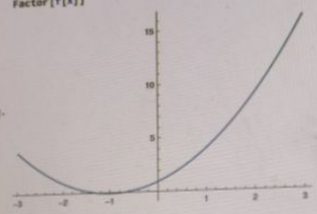
Out[•]=



Out[•]= Solve[{0, 0}, {x, y}]

```
In[•]:= f[x_] := (x^2+2x+1)
        Plot[f[x], {x, -3, 3}]
        Factor[f[x]]
```

Out[•]=



Out[•]= $(1+x)^2$

```
In[•]:= NSolve[x^2+2x+1 = 0, x]
```

Set: Tag Plus in $1+2x+x^2$ is Protected.

Out[•]= {{}}

```
In[•]:= CountryData["India", "Shape"]
```

In[3]:= {u1, u2, u3}.{v1, v2, v3}

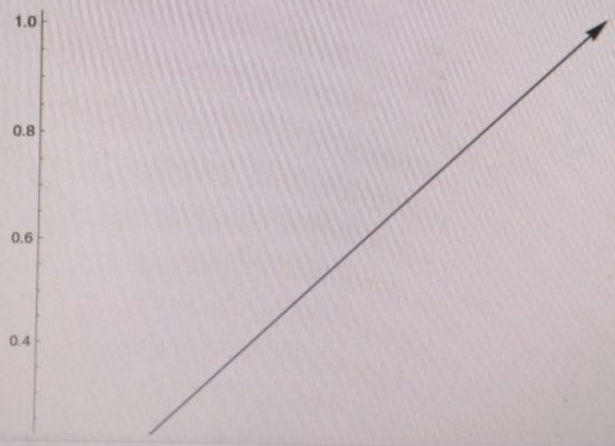
Out[3]= u1 v1 + u2 v2 + u3 v3

In[4]:= {a, b, c}.{2, 3, 4}

Out[4]= 2 a + 3 b + 4 c

In[6]:= Graphics[Arrow[{{θ, θ}, {1, 1}}], Axes → True]

Out[6]=

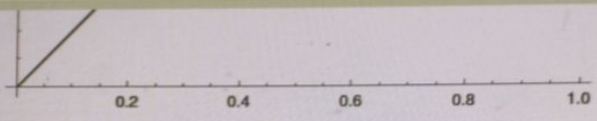


Activities Mathematica 13 Feb 16 11:42

MATHEMATICA.nb * - Wolfram M

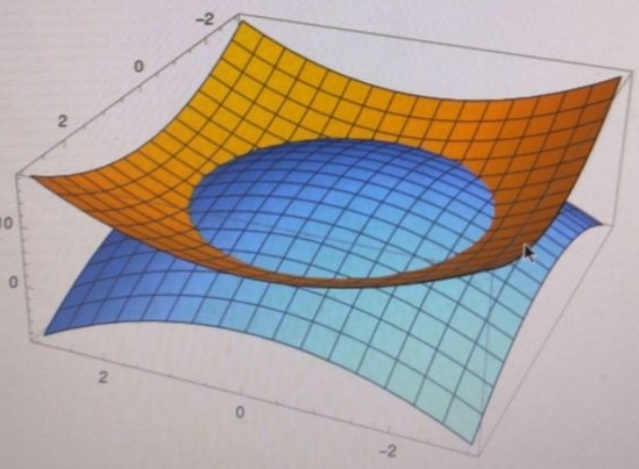
File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

Input



In[7]:= Plot3D[{x^2 + y^2, 9 - x^2 - y^2}, {x, -3, 3}, {y, -3, 3}]

Out[7]=

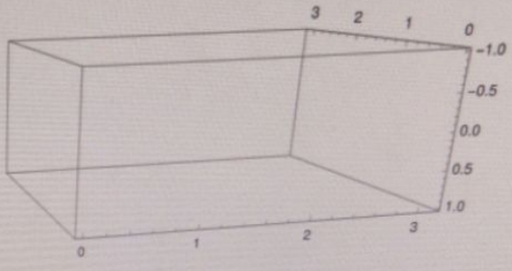


In[∘]:= Cross[{a, b, c}, {d, e, f}]

Out[∘]= {-c e + b f, c d - a f, -b d + a e}

In[∘]:= Plot3D[Sinx.Cosy, {x, θ, π}, {y, θ, π}]

Out[∘]=



θ

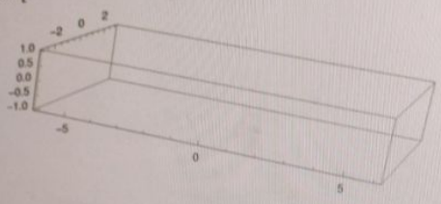
File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

In[•]:=
```
Cross[{a, b, c}, {d, e, f}]
```

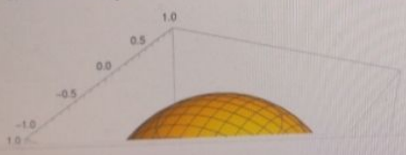
Out[•]=
```
{-c e + b f, c d - a f, -b d + a e}
```

In[•]:=
```
Plot3D[SinxCosy, {x, -6, 6}, {y, -3, 3}, BoxRatios → Automatic]
```

Out[•]=



In[•]:=
```
ContourPlot3D[x^2 + y^2 + z^2 == 1, {x, -1, 1}, {y, -1, 1}, {z, -1, 1}]
```

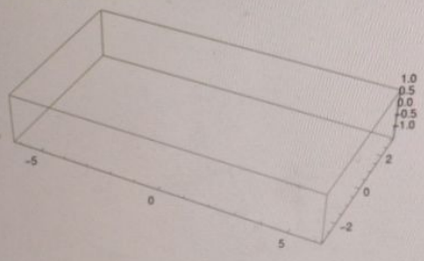


MATHEMATICA.nb - Wolfram Mathematica 13.2

File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

In[•]:=
```
Plot3D[Sin[XCos[Y]], {X, -6, 6}, {Y, -3, 3}, BoxRatios → Automatic]
```

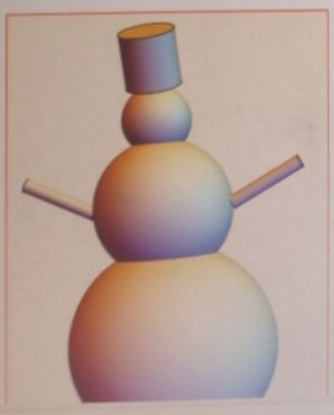
Out[•]=



In[•]:=
```
Graphics3D[{Sphere[{0, 0, 0}, 3], Sphere[{0, 0, 4}, 2], Sphere[{0, 0, 6.5}, 1],
  Cylinder[{{0, 0, 7.5}, {0, 0.5, 9}}, 0.8], Cylinder[{{0, 0, 7.5}, {0, 0.03, 7.6}}, 1.5],
  Cylinder[{{2, 0, 4}, {4, 0, 5}}, 0.2], Cylinder[{{-2, 0, 4}, {-4, 0, 5}}, 0.2]}, Boxed → False]
```

In[•]:=
```
Graphics3D[{Sphere[{0, 0, 0}, 3], Sphere[{0, 0, 4}, 2], Sphere[{0, 0, 6.5}, 1], Cylinder[{{0, 0, 7.5}, {0, 0.5, 9}}, 0.8],
  Cylinder[{{0, 0, 7, 5}, {0, 0.03, 7.6}}, 1.5], Cylinder[{{2, 0, 4}, {4, 0, 5}}, 0.2],
  Cylinder[{{-2, 0, 4}, {-4, 0, 5}}, 0.2]}, Boxed → False]
```

Out[•]=

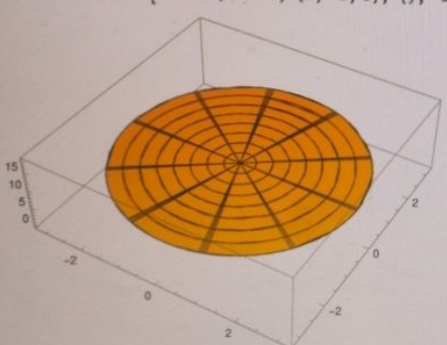


MATHEMATICA.nb * - Wolfram Mathematica 13.2

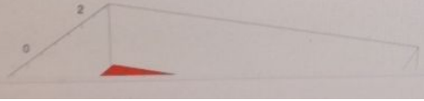
File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

```
RevolutionPlot3D[x^2 + 5 y^2 == 9, {x, -3, 3}, {y, -5, 5}]
```

Out[•]=



In[•]:=
```
Plot3D[x^(2/3) + y^(2/3) == 1, {x, -3, 3}, {y, -3, 3}, PlotStyle → Red]
```



In[∘]:= $\sum_{n=1}^{1\,000\,000\,000\,000\,953\,245\,364\,653\,764\,750\,000\,000\,000} n\wedge 2$

Sum: The variable n = 1 cannot be localized so that it can be assigned to numerical values.

Out[∘]= $\sum_{n=1}^{1\,000\,000\,000\,000\,953\,245\,364\,653\,764\,750\,000\,000\,000} n^2$

In[∘]:= {A1, A2, A3}.{B1, B2, B3}

Out[∘]= A1 B1 + A2 B2 + A3 B3

In[∘]:= $\prod_{n=0}^{5} (n)$

Out[∘]= 0

In[∘]:= Maximize[x^2 + y^2 + z^2, {x, y, z}]

Maximize: The maximum is not attained at any point satisfying the given constraints.

Out[∘]= $\{\infty, \{x \to -\infty, y \to -\frac{12}{5}, z \to -\frac{1}{2}\}\}$

MATHEMATICA.nb * - Wolfram Mathematica 13.2

In[∘]:= Minimize[x^2 + y^2 + z^2, {x, y, z}]

Out[∘]= {0, {x → 0, y → 0, z → 0}}

In[∘]:= $f = e^{\sin[x*y]} + \mathrm{Cos}\left[\frac{5x}{y^2+1}\right] * \mathrm{Sin}\left[\frac{3y}{x^2+1}\right]$

FindMaximum[f, {x, 1}, {y, -1}]

Out[∘]= $e^{\sin[x y]} + \mathrm{Cos}\left[\frac{5x}{1+y^2}\right] \mathrm{Sin}\left[\frac{3y}{1+x^2}\right]$

FindMaximum: The function value $-0.799137 - e^{\sin[-1]}$ is not a real number at {x, y} = {1., -1.}.

Out[∘]= FindMaximum[f, {x, 1}, {y, -1}]

In[∘]:= $\mathrm{ContourPlot}\left[e^{\sin[x*y]} + \mathrm{Cos}\left[\frac{5x}{1+y^2}\right] * \mathrm{Sin}\left[\frac{3y}{1+x^2}\right], \{x, -2, 2\}, \{y, -2, 2\}\right]$

MATHEMATICA.nb * - Wolfram Mathematica 13.2

In[∘]:= $\iiint (x\wedge 2 + y\wedge 2 + z\wedge 2)\, dx\, dy\, dz$

Out[∘]= $\frac{1}{3} x^3 y z + \frac{1}{3} x y^3 z + \frac{1}{3} x y z^3$

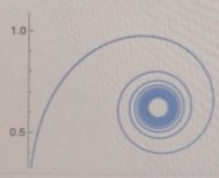
In[∘]:= $\int_0^1 \int_0^1 \int_0^1 (x\wedge 2 + y\wedge 2 + z\wedge 2)\, dx\, dy\, dz$

Out[∘]= 1

In[∘]:= $P[t_] = \left\{\int_0^t \mathrm{Sin}[u\wedge 2]\, du, \int_0^t \mathrm{Cos}[u\wedge 2]\, du\right\}$

ParametricPlot[P[t], {t, -10, 10}]

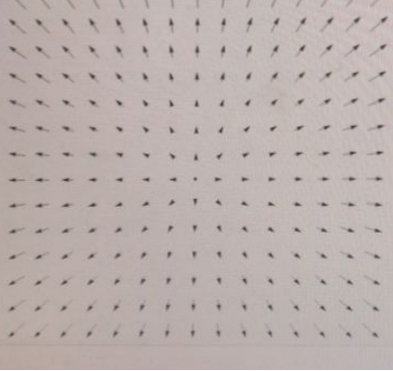
Out[∘]= $\left\{\sqrt{\frac{\pi}{2}}\, \mathrm{FresnelS}\left[\sqrt{\frac{2}{\pi}}\, t\right], \sqrt{\frac{\pi}{2}}\, \mathrm{FresnelC}\left[\sqrt{\frac{2}{\pi}}\, t\right]\right\}$



In[∘]:= Needs["VectorFieldPlots`"]
VectorFieldPlot[{x, y}, {x, -3, 3}, {y, -3, 3}]

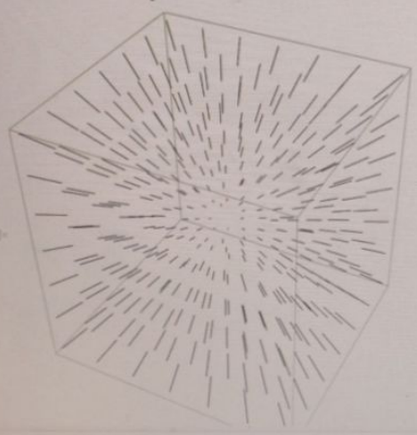
General: VectorFieldPlots` is now obsolete. The legacy version being loaded may conflict with current functionality. See the Compatibility Guide for updating information.

Out[∘]=



```
In[•]:= Needs["VectorFieldPlots`"]
        VectorFieldPlot3D[{x, y, z}, {x, -3, 3}, {y, -3, 3}, {z, -3, 3}]
```

Out[•]=



File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

Output

```
In[•]:= Mat1 = (1 2
                3 4)
Out[•]= {{1, 2}, {3, 4}}

In[•]:= Mat2 = {{1, 2}, {3, 4}}
        Mat2 // MatrixForm
Out[•]= {{1, 2}, {3, 4}}

Out[•]//MatrixForm=
        (1 2
         3 4)

In[•]:= Mat1 = (2 4
                6 8)
        Mat2 = (1 3
                5 7)
        3 Mat1 + 4 Mat2 // MatrixForm
Out[•]= {{2, 4}, {6, 8}}

Out[•]= {{1, 3}, {5, 7}}

Out[•]//MatrixForm=
        (10 24
```

File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

Output

```
In[•]:= 4 Mat1 - Mat2 // MatrixForm

Out[•]//MatrixForm=
        (7  13
         19 25)

In[•]:= Mat1 . Mat2 // MatrixForm

Out[•]//MatrixForm=
        (2  12
         30 56)

In[•]:= A = (2 3 5
             5 1 1
             1 2 3)
        B = (3
             5
             8)
        X = (x y z)
        AX = B
        LinearSolve[A, B]

Out[•]= {{2, 3, 5}, {5, 1, 1}, {1, 2, 3}}
```

MATHEMATICA.nb * - Wolfram Mathematica 13.2

File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

Output

```
In[•]:= A = (2 3 5
             5 1 1
             1 2 3)
        B = (3
             5
             8)
        X = (x y z)
        AX = B
        LinearSolve[A, B]

Out[•]= {{2, 3, 5}, {5, 1, 1}, {1, 2, 3}}

Out[•]= {{3}, {5}, {8}}

Out[•]= {{x, y, z}}

Out[•]= {{3}, {5}, {8}}

Out[•]= {{-8/5}, {147/5}, {-82/5}}
```

MATHEMATICA.nb * - Wolfram Mathematica 13.2

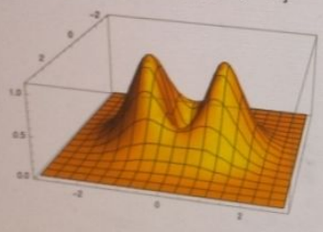
File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

Output

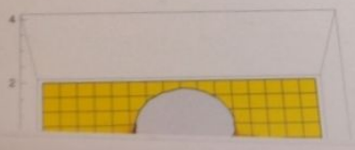
```
In[•]:= √2 + √3 + √5 + √8
Out[•]= 3 √2 + √3 + √5

In[•]:= N[π, 50]
Out[•]= 3.1415926535897932384626433832795028841971693993751

In[•]:= a = 5
        b = 3
        c = 8
        (a * b) - (a * c)
Out[•]= 5
Out[•]= 3
Out[•]= 8
Out[•]= -25

In[•]:= Z = (x^2 + 3 y^2) e^(-(x^2+y^2))
In[•]:= Plot3D[Z, {x, -3, 3}, {y, 3, -3}]
```

Output

```
In[•]:= Z = (x^2 + 3 y^2) e^(-(x^2+y^2))
In[•]:= Plot3D[Z, {x, -3, 3}, {y, 3, -3}]
Out[•]=
```




```
In[•]:= Plot3D[Z, {x, -4, 4}, {y, 4, -4}]
```



MATHEMATICA.nb * - Wolfram Mathematica 13.2

File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

Output

```
In[•]:= 180° → R
        Degree // N
Out[•]= 1 → R
Out[•]= 0.0174533

In[•]:= 8!
Out[•]= 40320

In[•]:= Random[Real]
Out[•]= 0.461725

In[•]:= Random[Complex]
Out[•]= 0.449174 + 0.887178 i

In[•]:= Random[Real, {1, 2}]
Out[•]= 1.81508

In[•]:= Random[Integer, {2, 4}]
```

File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

Output

```
In[•]:= Random[Integer, {2, 4}]
Out[•]= 4

In[•]:= Prime[1]
Out[•]= 2

In[•]:= Prime[2]
Out[•]= 3

In[•]:= Prime[100]
Out[•]= 541

In[•]:= Prime[3]
Out[•]= 5

In[•]:= Prime[4]
Out[•]= 7

In[•]:= Fibonacci[1]
Out[•]= 1
```

File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

Output

```
In[•]:= Fibonacci[10]
Out[•]= 55

In[•]:= IntegerPart[3.578]
Out[•]= 3

In[•]:= IntegerPart[10/3]
Out[•]= 3

In[•]:= FractionalPart[3.578]
Out[•]= 0.578

In[•]:= FractionalPart[Sqrt[2]]
Out[•]= -1 + Sqrt[2]

In[•]:= GCD[10, 15]
Out[•]= 5

In[•]:= LCM[10, 15]
```

File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

Output

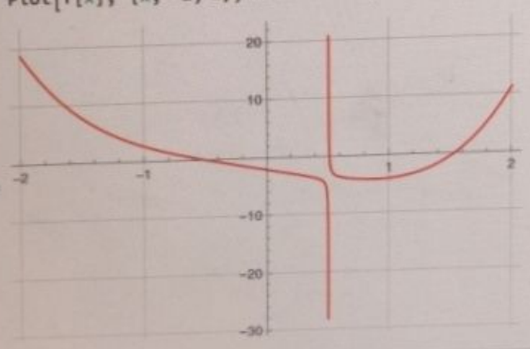
```
In[•]:= LCM[5, 10, 15, 20, 25, 30, 35, 40, 45]
Out[•]= 12600

In[•]:= GCD[3, 8, 13, 26, 34]
Out[•]= 1

In[•]:= LCM[3, 8, 13, 26, 34]
Out[•]= 5304

In[•]:= Factor[583920]
Out[•]= 583920

In[•]:= N[Log2 1000]
Out[•]= 1000. Log2

In[11]:= f[x_] := (x^5 - 4 x^2 + 1)/(x - 1/2)
         Plot[f[x], {x, -2, 2}, Axes → True, GridLines → Automatic, PlotStyle → Red]
```

Activities Mathematica 13 Mar 21 13:23

MATHEMATICA.nb * - Wolfram Mathematica 13.2

File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

Insert Cell...

```
In[•]:= N[Log2 1000]
Out[•]= 1000. Log2

In[11]:= f[x_] := (x^5 - 4 x^2 + 1)/(x - 1/2)
         Plot[f[x], {x, -2, 2}, Axes → True, GridLines → Automatic, PlotStyle → Red]
```

Out[12]=